

Large Dual-Lumen Extracorporeal Membrane Oxygenation Cannulas Are Associated with More Intracranial Hemorrhage

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Large dual-lumen veno-venous (VV) extracorporeal membrane oxygenation (ECMO) cannulas may increase venous pressure in the brain, contributing to intracranial hemorrhage (ICH). A retrospective cohort study was performed using the extracorporeal life support organization (ELSO) registry. Propensity score matching was used to control for confounding. The rate of ICH and rates of hemolysis, cannula complications, and mortality were compared between patients with a 27 French dual-lumen cannula and patients with a 31 French dual-lumen cannula. Seven hundred forty-four patients were included in the propensity score-matched cohort. Patients were well matched except there was some residual imbalance in body weight and sex. Patients with a 31 French cannula had an ICH rate of 4.3% compared with 1.6% in patients with a 27 French cannula ($p = 0.03$). There were no differences in hemolysis, cannula complications, or mortality between groups. After controlling for body weight and sex in the matched cohort, ICH odds remained higher in patients with a 31 French cannula; odds ratio = 2.74, (95% CI = 1.06–7.09, $p = 0.03$). Our study data suggest that large dual-lumen VV ECMO cannulas are associated with more ICH, and thus, smaller cannulas may be preferable when feasible. *ASAIO Journal* 2019; 65:674–677.

Key Words: cannulas, intracranial hemorrhage

Dual-lumen veno-venous (VV) extracorporeal membrane oxygenation (ECMO) cannulas are used for adult patients with respiratory failure in many ECMO centers and have several potential advantages over conventional two site cannulation including easier ambulation for patients, reduced sedation requirements, and earlier extubation.¹ Dual-lumen cannulas are also associated with serious complications related to their size and design including bleeding, migration into the right ventricle, and even right ventricular rupture.^{2,3}

Patients with dual-lumen VV ECMO cannulas have a high rate of intracranial hemorrhage (ICH). In one cohort of 72

patients with a dual-lumen VV ECMO cannula (51% 27 French and 49% 31 French), the rate of ICH was 7%.⁴ There are a number of factors that likely contribute to this high ICH rate including thrombocytopenia during ECMO, anticoagulation, and possibly intracranial venous hypertension from obstruction of the internal jugular vein and superior vena cava by the ECMO cannula. Further, a large number of patients develop cannula-associated deep venous thrombosis (DVT) during ECMO, which may contribute to intracranial venous hypertension.⁵

Two of the most commonly used dual-lumen VV ECMO cannula sizes in adult patients are 27 and 31 French. At the present time, it remains unclear which cannula is optimal for the “average” adult VV ECMO patient. In fact, there is a trade-off when considering which cannula size to use because the 31 French cannula allows for greater ECMO blood flow, which improves oxygenation, but is also potentially associated with more cannula complications and a higher rate of cannula site bleeding.^{2,4} To our knowledge, there are no clinical studies comparing outcomes between the two dual-lumen cannula sizes. We hypothesized that 31 French dual-lumen VV ECMO cannulas would be associated with a higher rate of ICH than 27 French cannulas given their larger size.

Methods

Subjects

The extracorporeal life support organization (ELSO) registry was queried to identify all adult patients who had VV ECMO between 2011 and 2016 and had either a 27 or 31 French dual-lumen VV ECMO cannula. Extracorporeal life support organization was not responsible for our data analysis or interpretation. Informed consent was waived as the study was not human subjects research and a deidentified dataset was used for the analysis. Patients who had a body weight above 150 kg or below 50 kg were excluded from the analysis.

Variables

For all patients, we collected the following variables, which were available in the ELSO database: age, sex, body weight, cannula size, pre-ECMO arrest, FiO₂ before ECMO, peak inspiratory pressure before ECMO, positive end expiratory pressure before ECMO, pH before ECMO, PaCO₂ before ECMO, PaO₂ before ECMO, intubation hours before ECMO, ECMO blood flow at 4 and 24 hours after ECMO initiation, and total ECMO hours.

Outcomes

The study's primary outcome variable was ICH, as defined in the ELSO registry. Secondary outcomes were significant

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Table 1. Patient Characteristics in Propensity Score-Matched Cohort

Variable	27 French N = 372	31 French N = 372	SD
Age	45 ± 16	46 ± 14	0.07
Weight (kg)	78 ± 19	82 ± 18	0.22
Sex (% male)	150 (40.3)	193 (51.9)	0.23
Pre-ECMO arrest	24 (6.5)	23 (6.2)	0.01
Fio2 (% oxygen)	1.0 [1.0–1.0]	1.0 [1.0–1.0]	<0.01
Peak inspiratory pressure	35 [30–40]	35 [30–40]	<0.01
PEEP (cmH2O)	14 [10–16]	14 [10–17]	0.02
pH before ECMO	7.24 [7.14–7.35]	7.25 [7.15–7.34]	0.06
PCO2 (mmHg) before ECMO	56 [43–72]	54 [42–73]	<0.01
PO2 (mmHg) before ECMO	58 [44–71]	56 [44–71]	0.08
Intubation hours before ECMO	48 [17–133]	48 [16–140]	0.03
Pump flow (L/min) 4 hours	4.2 [3.7–4.7]	4.2 [3.6–4.8]	0.03
Pump flow (L/min) 24 hours	4.2 [3.8–4.8]	4.3 [3.8–4.9]	0.08
Total ECMO hours	166 [87–354]	184 [93–363]	0.03

Values are median [interquartile range] or number (% of patients)

ECMO, extracorporeal membrane oxygenation; PEEP, positive end expiratory pressure; SD, standardized difference.

hemolysis, cannula complications, and in-hospital mortality. Significant hemolysis is defined in the ELSO registry as plasma free hemoglobin level above 50 mg/dL.

Sample Size Calculation

Assuming a 7% rate of ICH in the high-risk group, an alpha of 5%, and 80% power, a total of 317 patients were needed in each group to detect a 50% lower ICH rate in the low-risk group.

Statistical Analysis

Statistical analysis was performed using SAS 9.3 (SAS Corp, Cary, NC). A propensity score model was created using logistic regression with the dependent variable being receipt of a 27 French cannula and all the previously mentioned patient and ECMO variables entered into the model as independent variables. After propensity scores were calculated for each patient, global optimal matching was used to create a propensity score-matched cohort. Standardized differences were calculated to assess the balance of covariates in the matched cohort.

Both unadjusted and adjusted odds ratios with 95% CIs were calculated for the primary outcome variable and for secondary outcome variables. Adjusted odds ratios controlled for variables that could not be fully balanced with the propensity score-matching process.

Results

The propensity score-matched cohort included 744 patients. Patient characteristics for both groups with standardized differences are shown in **Table 1**. All covariates were well balanced between the two groups except for sex and body weight, which had some residual imbalance.

Table 2 shows study outcomes by cannula size group. Patients with a 31 French cannula had an approximately three-fold higher rate of ICH (4.3% vs. 1.6%, $p = 0.03$). There were no significant differences in the rates of hemolysis, cannula complications, or mortality between the two groups (all $p > 0.05$). **Table 3** shows both unadjusted and adjusted (for body weight and sex) odds ratios for study outcomes. The adjusted

odds ratio (OR) for ICH with a 31 French cannula remained significant; OR = 2.83 (95% CI = 1.08–7.42) after adjusting for body weight and sex, but no other adjusted odds ratio was significant.

Discussion

Neurologic complications occur in 7% of patients on VV ECMO with ICH representing the most common form of injury (43%), followed by brain death (24%), and ischemic stroke (20.0%).⁶ Intracranial hemorrhage is reported to occur in as many as 20% of adult ECMO patients, depending on the center, and is associated with a variety of risk factors including high precannulation sequential organ failure assessment (SOFA) coagulation score, low platelet count, and spontaneous extracranial hemorrhage.⁷ The mortality rate for patients with ICH during ECMO is 81% at 1 month and 85% at 6 months, reflecting its devastating nature.⁷ To our knowledge, no prior studies have evaluated whether cannulation strategy affects the rate of ICH in adult patients on VV ECMO. Cannulation strategy has the potential to impact ICH rates because dual-lumen VV ECMO cannulas, 27–31 French, are larger than two site return cannulas, which are usually 19–23 French for adult patients.

Dual-lumen VV ECMO cannulas can be used in most adult patients who require VV ECMO and are associated with relatively low device failure and thrombosis rates.^{8,9} Dual-lumen VV ECMO cannulas are also associated with a relatively high complication rate of up to 38%. The most frequently reported complication is cannula site bleeding (20%), followed by cannula site infection (10%), and ICH (7%).⁴

Table 2. Patient Outcomes

Variable	27 French N = 372	31 French N = 372	<i>p</i> Value
Intracranial hemorrhage	6 (1.6)	16 (4.3)	0.03
Hemolysis	17 (4.6)	24 (6.5)	0.26
Cannula complication	23 (6.2)	21 (5.6)	0.76
In-hospital mortality	145 (39.0)	145 (39.0)	1.0

values are number (% of patients).

Table 3. Adjusted and Unadjusted Odds Ratios for Outcomes with 31 French versus 27 French Cannula

	OR	95% CI	p Value
ICH			
Unadjusted	2.74	1.06–7.09	0.03
Adjusted for sex and weight	2.83	1.08–7.42	0.03
Hemolysis			
Unadjusted	1.44	0.76–2.73	0.26
Adjusted for sex and weight	1.45	0.76–2.77	0.26
Cannula complication			
Unadjusted	0.91	0.49–1.67	0.76
Adjusted for sex and weight	0.99	0.53–1.84	0.97
Mortality			
Unadjusted	1.0	0.75–1.34	1.0
Adjusted for sex and weight	0.99	0.74–1.34	0.97

ICH, intracranial hemorrhage; OR, odds ratio.

The pathophysiology of ICH during VV ECMO is not fully understood but is likely multifactorial and is related to platelet dysfunction, systemic anticoagulation, abnormal production of procoagulant factors during shock, loss of large von willebrand factor (VWF) multimers, and intracranial venous hypertension from DVT or cannula-related obstruction of the internal jugular veins.^{10–12} Bleeding complications are common in adult ECMO patients occurring in up to 50% of patients.¹³ Cannula-associated DVT is also common occurring in up to 85% of ECMO patients.⁵ There are no animal studies confirming that VV ECMO cannulas cause venous congestion in the brain, but it is reasonable to assume that they increase cerebral venous pressure, as this phenomenon has been reported in patients with much smaller central venous catheters (8 French), particularly in cases of bilateral internal jugular vein cannulation.¹⁴ Based on these observations, we hypothesized that large dual-lumen VV ECMO cannulas would increase the risk for ICH, and in fact, our study data confirmed this.

Dual-lumen cannulas (27 French) can achieve maximum blood flows of approximately 4.5 L/min, whereas 31 French cannulas can achieve blood flows up to 6 L/min.¹⁵ Because oxygenation is directly related to ECMO blood flow, a 31 French cannula offers a theoretical advantage in patients with severely impaired oxygenation. However, our analysis did not suggest any benefit from using a 31 French cannula. In fact, patients in our cohort had comparable severity of pre-ECMO lung injury, and ECMO blood flow at 4 and 24 hours was equivalent. In addition, there was no difference in hemolysis between the two groups. These data suggest that adequate ECMO blood flow can be achieved in an “average” patient using either cannula without having to utilize excessively high revolutions per minute (RPMs). A 27 French cannula may be preferable though because there appears to be less ICH.

Our study has several critical limitations. First, it was observational, and data were pooled in the ELSO registry from multiple participating centers with highly variable practices. Second, although patients with a 31 French cannula had more ICH, they had the same mortality as patients with a 27 French cannula. Unfortunately, the ELSO database does not have data on quality of life or functional outcome so it is difficult to discern whether there was a difference in functional survival between the two groups. Third, although we used propensity score matching to balance covariates between the groups, we cannot rule out the possibility of residual confounding which could have biased our

results. In fact, we did not include some previously described risk factors for ICH in our propensity score analysis because these data were not readily available in the ELSO registry (e.g., daily platelet counts, degree of anticoagulation, type of anticoagulation, and international normalized ratio values). Finally, in our cohort, the rates of ICH were 1% and 4% in the two groups. This is quite different from ICH rates reported in other studies, where ICH was as high as 20%.⁶ Our rate of ICH is similar to that of the conventional ventilatory support versus extracorporeal membrane oxygenation for severe adult respiratory failure (CESAR) trial, but we cannot rule out the possibility of ICH under-reporting in the ELSO registry.¹⁶

In summary, our data suggest that 31 French dual-lumen VV ECMO cannulas are associated with more ICH than 27 French cannulas. There was no difference in mortality, hemolysis, cannula complications, or ECMO blood flow at 4 and 24 hours between groups. Given these data, a 27 French cannula may be preferable to a 31 French dual-lumen cannula for most adult VV ECMO patients.

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